Assignment No. - 7

Title: Stored Procedures or Function with Cursors.

Problem Statement:
Create and execute stored procedures / function using cursors.

Objective:

- ***To learn procedure and function in Pl/SQL.***
- *To learn concept of **cursor and** types of **cursor.***
- ***To implement** cursor in PL/SQL.*

Theory: CURSOR:-

For the processing of any SQL statement, database needs to allocate memory. This memory is calledcontext area. The context area is a part of PGA (Process global area) and is allocated on the oracle server. A cursor is associated with this work area used by ORACLE, for multi row queries. A cursor is a handleor pointer to the context area .The cursor allows to process contentsin the context area row by row.

THERE ARE TWO TYPES OF CURSORS.

Implicit cursor:-Implicit cursors are defined by ORACLE implicitly. ORACLE defines implicitcursor for every DML statements.

Explicit cursor:-These are user-defined cursors which are defined in the declaration section of thePL/SQL block. There are four steps in which the explicit cursor is processed.

- Declaring a cursor
- Opening a cursor
- Fetching rows from an opened cursor
- Closing cursor

General syntax for CURSOR:- DECLARE

Cursor cursor_name IS select_statement or query;BEGIN Open cursor_name;
FETCH CURSOR_NAME INTO LIST_OF_VARIABLES;CLOSE CURSOR_NAME;END;
Where

PL/SQL Cursor :

When an SQL statement is processed, Oracle creates a memory area known as context area.
A cursor is a pointer to this context area. It contains all information needed for processing

the statement. In PL/SQL, the context area is controlled by Cursor. A cursor contains
information on a select statement and the rows of data accessed by it. A cursor is used to referred to a program to fetch and process the rows returned by the SQL statement, one at a time. There are two types of cursors:

- Implicit Cursors
- Explicit Cursors

1. PL/SQL Implicit Cursors

The implicit cursors are automatically generated by Oracle while an SQL statement is executed, if you don't use an explicit cursor for the statement. These are created by default to process the statements when DML statements like INSERT, UPDATE, DELETE etc. are executed.Orcale provides some attributes known as Implicit cursor's attributes to check the status of DML operations. Some of them are: %FOUND, %NOTFOUND, %ROWCOUNT and %ISOPEN.

For example: When you execute the SQL statements like INSERT, UPDATE, DELETE then the
cursor attributes tell whether any rows are affected and how many have been affected. If you run a SELECT INTO statement in PL/SQL block, the implicit cursor attribute can be used to find out whether any row has been returned by the SELECT statement. It will return an error if there no data is selected.The following table soecifies the status of the cursor with each of its attribute.

PL/SQL Implicit Cursor Example

Create customers table and have records:

ID	NAME	AGE	ADDRESS	SALARY
1	Ramesh	23	Allahabad	20000
2	Suresh	22	Kanpur	22000
3	Mahesh	24	Ghaziabad	24000
4	Chandan	25	Noida	26000
5	Alex	21	Paris	28000
6	Sunita	20	Delhi	30000

Let's execute the following program to update the table and increase salary of each customer by 5000. Here, SQL%ROWCOUNT attribute is used to determine the number of rows affected: is procedure in SQL , total_rows number(2);

```
BEGIN
UPDATE  customers  SET salary = salary + 5000;
IF sql%notfound  THEN dbms_output.put_line('no customers
  updated');
ELSIF sql%found THEN total_rows := sql%rowcount;
dbms_output.put_line( total_rows || ' customers updated ');
```

```
  END IF;
END;
/
```

Output:

6 customers updated
PL/SQL procedure successfully completed.

Now, if you check the records in customer table, you will find that the rows are updated.

select * **from** customers;

ID	NAME	AGE	ADDRESS	SALARY
1	Ramesh	23	Allahabad	25000
2	Suresh	22	Kanpur	27000
3	Mahesh	24	Ghaziabad	29000
4	Chandan	25	Noida	31000
5	Alex	21	Paris	33000
6	Sunita	20	Delhi	35000

PL/SQL Explicit Cursors:

The Explicit cursors are defined by the programmers to gain more control over the contextarea.
These cursors should be defined in the declaration section of the PL/SQL block. It is created on a SELECT statement which returns more than one row.
Following is the syntax to create an explicit cursor: Syntax of explicit cursor
Following is the syntax to create an explicit cursor: **CURSOR** cursor_name **IS** select_statement;; **Steps:`**
You must follow these steps while working with an explicit cursor.

- Declare the cursor to initialize in the memory.
- Open the cursor to allocate memory.
- Fetch the cursor to retrieve data.
- Close the cursor to release allocated memory.

Declare the cursor:

It defines the cursor with a name and the associated SELECT statement.

Syntax for explicit cursor decleration CURSOR name IS

SELECT statement;

1. Open the cursor:

It is used to allocate memory for the cursor and make it easy to fetch the rows returned by theSQL statements into it.

Syntax for cursor open:

```
OPEN cursor_name;
```

2. Fetch the cursor:
3.

It is used to access one row at a time. You can fetch rows from the above-opened cursor asfollows:

Syntax for cursor fetch:

FETCH cursor_name **INTO** variable_list;

4. Close the cursor:

It is used to release the allocated memory. The following syntax is used to close the above-opened cursors.

Syntax for cursor close:

Close cursor_name;

PL/SQL Explicit Cursor Example

Explicit cursors are defined by programmers to gain more control over the context area. It is definedin the declaration section of the PL/SQL block. It is created on a SELECT statement whichreturns more than one row.

Let's take an example to demonstrate the use of explicit cursor. In this example, we are using the already created CUSTOMERS table.

Create customers table and have records:

ID	NAME	AGE	ADDRESS	SALARY
1	Ramesh	23	Allahabad	20000
2	Suresh	22	Kanpur	22000
3	Mahesh	24	Ghaziabad	24000
4	Chandan	25	Noida	26000
5	Alex	21	Paris	28000
6	Sunita	20	Delhi	30000

Create procedure:

Execute the following program to retrieve the customer name and address.

```
DECLARE
    c_id  customers.id%type; c_name customers.name%type;
    c_addr customers.address%type;
  CURSOR c_customers is
    SELECT id, name, address FROM customers;
BEGIN
```

```
OPEN c_customers;

        LOOP
        FETCH
        c_customers
        into        c_id,
        c_name,
        c_addr;   EXIT
        WHEN
        c_customers%
        notfound;
        dbms_output.put_line(c_id || ' ' || c_name || ' ' || c_addr);
      END LOOP;
      CLOSE c_customers;
    END; /
```

Output:

i. Ramesh Allahabad
ii. Suresh Kanpur
iii. Mahesh Ghaziabad
iv. Chandan Noida
v. Alex Paris
vi. Sunita Delhi

DRAWBACKS OF IMPLICIT CURSORS

Even if your query returns only a single row, you might still decide to use an explicit cursor. Theimplicit cursor has the following drawbacks:
It is less efficient than an explicit cursor

1. It is more vulnerable to data errors
2. It gives you less programmatic control

The following sections explore each of these limitations to the implicit cursor.

Conclusion:
Thus we have studied the concept of cursor in PL/SQL procedure / function and implemented in SQL.

FAQ:

1. What is procedure in SQL?
2. What is function in SQL?
3. Explain the cursor operation.
4. What are types of cursors?